SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 20%

NAME: M.NITHIYASHREE
REG. NO: 192421101

Scenario-Based Questions – Image Processing

1. Salt-and-Pepper Noise

(a) Scenario Interpretation

A scanned document contains random black and white pixels (salt-and-pepper noise), making the text difficult to read.

(b) Key Issues

- Reduces image clarity.
- Affects readability of text.
- Introduces unwanted black and white spots.
- Makes character recognition difficult.

(c) Appropriate Filtering Technique

Median Filter

(d) Justification

The median filter is the most suitable because it effectively removes salt-and-pepper noise while preserving edges. Unlike the averaging filter, it does not blur the image significantly.

(e) Conclusion

Median filtering improves document readability by removing impulse noise and preserving important image details.

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2. Object Boundary Detection

(a) Scenario Interpretation

In medical images, tissue boundaries must be detected accurately for diagnosis and analysis.

(b) Key Challenges

- Low image contrast.
- Presence of noise.
- Weak or blurred edges.
- Similar intensity values between tissues.

(c) Suitable Edge Detection Method

Sobel Edge Detection

(d) Justification

The Sobel operator detects horizontal and vertical edges while reducing the effect of noise. It is simple, efficient, and widely used in medical image processing.

(e) Conclusion

Sobel edge detection accurately highlights tissue boundaries, making medical image analysis easier.

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3. Simple Segmentation

(a) Scenario Interpretation

The objects are clearly different from the background based on intensity values.

(b) Key Features

- High contrast between object and background.
- Uniform object intensity.
- Distinct pixel values.

(c) Suitable Segmentation Technique

Global Thresholding

(d) Justification

Thresholding separates foreground and background using a single threshold value. It is simple, fast, and effective when the object intensity differs significantly from the background.

(e) Conclusion

Thresholding accurately separates the objects from the background with minimum computational complexity.

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4. Low Brightness Image

(a) Scenario Interpretation

The image appears dark because it was captured under poor lighting conditions.

(b) Key Issues

- Poor visibility.
- Low contrast.
- Loss of image details.
- Difficult object recognition.

(c) Appropriate Enhancement Technique

Histogram Equalization (or Log Transformation)

(d) Justification

Histogram equalization redistributes pixel intensities over the full gray-level range, improving brightness and contrast. It enhances dark regions without affecting image quality significantly.

(e) Conclusion

Histogram equalization improves image visibility and makes hidden details easier to observe.

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5. Gaussian Noise

(a) Scenario Interpretation

The image contains Gaussian noise, which appears as random intensity variations throughout the image.

(b) Key Issues

- Reduces image quality.
- Decreases visual clarity.
- Makes feature extraction difficult.
- Affects smooth regions.

(c) Appropriate Smoothing Filter

Gaussian Filter

(d) Justification

The Gaussian filter reduces Gaussian noise while preserving edges better than the averaging filter. It assigns higher weight to nearby pixels, producing smoother results.

(e) Conclusion

Gaussian filtering effectively removes Gaussian noise and improves overall image quality.

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6. Loss of Fine Details

(a) Scenario Interpretation

The image loses fine details after excessive smoothing or filtering.

(b) Key Issues

- Blurred edges.
- Loss of texture.
- Reduced sharpness.
- Poor feature visibility.

(c) Suitable Enhancement Technique

Laplacian Sharpening Filter (or Unsharp Masking)

(d) Justification

Laplacian sharpening enhances edges and restores fine image details by emphasizing intensity changes. It improves image sharpness after smoothing.

(e) Conclusion

Sharpening restores lost details and improves the overall appearance of the image.

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7. Edge Enhancement Requirement

(a) Scenario Interpretation

The application requires clear edge information for feature extraction and object recognition.

(b) Key Issues

- Weak edges.
- Image noise.
- Low contrast.
- Difficulty identifying object boundaries.

(c) Appropriate Method

Sobel Edge Detection (or Canny Edge Detection)

(d) Justification

The Sobel operator detects edge gradients efficiently and is computationally simple. For higher accuracy and better noise suppression, the Canny edge detector can also be used.

(e) Conclusion

Edge enhancement improves feature extraction by making object boundaries clearer and more distinguishable.